

Breaking human dominance: Investigating learners' preferences for learning feedback from generative AI and human tutors

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Abstract

Understanding learners' preferences in educational settings is crucial for optimizing learning outcomes and experience. As artificial intelligence (AI) becomes increasingly integrated into educational contexts, it is crucial to understand learners' preferences between AI and human tutors to support their learning. While AI demonstrates growing potential in education, the phenomenon of algorithm aversion, which is a tendency to favour human decision making over algorithmic solutions, requires further investigation. To explore this issue, an experiment involving 114 university students was conducted to measure learners' preferences for different feedback sources before and after exposure to one of four conditions: no feedback, human tutor feedback, ChatGPT feedback through a free-dialogue user interface, and AI-powered writing analytics tool feedback through a structured interface. Our results revealed a strong initial preference for human tutors. However, the post-task analysis showed an important nuance. While the general preference for human tutors persisted, learners' preference towards the free-dialogue interface (ChatGPT 4.0) of ChatGPT increased, whereas the structured AI interface (AI-powered writing analytics tool) reinforced the preference for human tutors. These findings offer theoretical and practical contributions by extending algorithm aversion theory to

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educational contexts and demonstrating that appropriate interaction design can mitigate this aversion. The success of free-dialogue interfaces suggests that overcoming algorithm aversion may depend more on creating natural, flexible interaction experiences than purely technical optimization. However, we must also consider that increased preference for AI tools, particularly those with more engaging interfaces, may potentially lead to over-reliance and metacognitive laziness among learners, highlighting the importance of balancing technological support with the development of independent learning skills.

KEYWORDS

AI in education, algorithm aversion, feedback preference, large language models

Practitioner notes

What is already known about this topic?

- Algorithm aversion exists across various contexts where individuals tend to prefer human over algorithmic decision-making.
- The introduction of generative AI brings new possibilities for AI-supported learning.

What this paper adds?

- In academic writing tasks, learners show strong initial preference for human tutors over Generative AI feedback.
- Strong initial preference for human tutors persists even after exposure to generative AI feedback.
- Different interaction designs lead to divergent preference patterns: Free-dialogue interface increases preference for AI feedback, structured interface reinforces preference for human tutors.

Implications for practice and/or policy

- Algorithm aversion in educational contexts can be mitigated through appropriate interaction design, particularly through natural dialogue interfaces.
- Design AI educational tools with back-and-forth, conversational interfaces to reduce algorithm aversion.

INTRODUCTION

Since the 1960s, researchers have envisioned that computers could enhance human learning by providing personalized support and feedback (Carbonell, 1970). More than half a century later, advancements in technologies, such as generative have brought this vision closer to realization. As we move towards a future where humans might work more closely with AI in educational settings, studies from human–computer interaction raise an important concern that users often demonstrate a systematic preference for feedback from humans over algorithms (Dietvorst et al., 2015; Jussupow et al., 2020). Even when

AI demonstrates superior performance on specific tasks, users often prefer human expert judgement over AI-generated recommendations (Eastwood et al., 2012). This phenomenon is called *algorithm aversion*. As defined by Jussupow et al. (2020), algorithm aversion is *a biased assessment of an algorithm that manifests itself in negative behaviours and attitudes towards the algorithm compared to a human agent*. Existing studies indicate that algorithm aversion has a robust effect on individuals' decision-making processes in multiple contexts. For example, patients are more inclined to medical solutions provided by human doctors instead of AI algorithms (Longoni et al., 2019). In economics, people often place greater value on suggestions from human experts compared to those provided by algorithmic models (Önköl et al., 2009). These findings suggest a complex psychological phenomenon where individuals demonstrate a persistent preference (Jussupow et al., 2020), and this preference appears to be driven by various psychological factors beyond pure performance considerations.

In educational settings, although there is evidence from multiple systematic reviews and meta-analyses showing that AI applications outperform human teachers on various educational tasks (Kulik & Fletcher, 2016; VanLehn, 2011), learners show scepticism towards AI-provided feedback over feedback provided by human tutors. We posit that this scepticism may stem from algorithm aversion, especially in educational contexts involving intensive and continuous decision making. From the perspective of self-regulated learning, learners make decisions throughout the process of setting goals, selecting strategies and engaging in self-reflection (Pintrich, 1991, 2000). As such, the "inherent aversion" to AI feedback—a tendency to favour human over algorithmic decision making even when algorithms demonstrate superior performance (Jussupow et al., 2020)—might also manifest in teaching and learning environments.

The prevalence of algorithm aversion presents a significant challenge for the implementation of AI in education. When learners show a preference for human feedback over AI feedback, it may undermine the effectiveness of AI applications, regardless of their technical capabilities. For example, learners may disengage from AI feedback due to a stronger preference for human tutors. Learners may also distrust AI feedback, which can result in underutilization of AI suggestions and hinder their ability to adapt learning strategies to better align with the AI application (Hsu et al., 2021). Furthermore, the reluctance to rely on feedback from AI might increase cognitive load, as students spend more time doubting AI capabilities rather than focusing on learning (Le & Jia, 2022).

Given this challenging outlook for the application of AI in gaining learner preference for receiving learning feedback, the emergence of generative AI presents an opportunity for change. Generative AI, such as ChatGPT, has sparked new opportunities for providing learning feedback, particularly due to their cutting-edge performance in natural language processing (Koraishi, 2023; Nguyen, 2023; Qureshi, 2023; Zehnle, 2025). However, a critical question remains: can AI applications in education effectively address the strong preference for human feedback, given that this preference appears to be deeply rooted in psychological and social factors? Beyond the technical capabilities of AI systems, the way these systems interact with learners may also play a crucial role in shaping learners' preferences. These interactions take shape through two distinct modes in current AI applications in education: free-form dialogue interfaces that enable natural conversations (as seen in ChatGPT) and structured interfaces that provide relatively static feedback. These different approaches to human–AI interaction may significantly influence how learners perceive and engage with AI support, potentially shaping their preferences between human and AI support in education.

To address these issues, the current study systematically investigated learners' preferences for different feedback sources—human tutors, AI in free-dialogue and AI-powered writing analytics tool—in the context of academic writing tasks, aiming to understand how

different feedback sources and interaction modes influence learners' preferences. The study offers both theoretical and practical contributions to the field. Theoretically, it enhances our understanding of algorithm aversion in educational contexts and sheds some light on how interaction design might moderate this effect. Practically, our findings can inform the design of more effective AI educational systems that better align with learner preferences.

BACKGROUND

Learning preference for educational feedback

Preference is a fundamental concept that arises in contexts where choices are available. Initially discussed in economics, preferences are expressed through choices and decisions to facilitate decision making. Warren et al. (2011) noted that there are two ways of defining preferences. One prevalent in economics involves rank order choices (e.g., consumers select A over B and C), noted as 'expressed preferences'. This is a traditional way of operationalizing preferences. In psychology, the term preference denotes a latent tendency to regard something desirable or undesirable, suggesting that preferences can be operationalized as scale ratings or measure attitudes, often referred to as 'underlying preferences' (Tversky et al., 1988). Accordingly, there are generally two approaches to measure preference: The first one involves providing individuals with assortments and letting them rank an order of the given choices, while the second uses scales (Warren et al., 2011).

Learning preference refers to students' choice of one learning situation over another (Rezler & Rezmovic, 1981). This concept can be used to identify learners' needs and provide them with personalized content (Zhang et al., 2007). The word 'learning preference' is developed in educational psychology. Its early usage could be dated back to the 1960s (James, 1962). Learning preference is taken as an inherently apparent construct that could potentially facilitate learning. Scholars have held the belief that learners, as proactive agents, would learn better in a preferred manner (Bonk et al., 2015; James, 1962; Lynch et al., 2001; Reid, 1987; Zhang et al., 2021). However, the research since then has shown important limitations in its applications in education in connection to the widely debunked concept of learning styles. Notably, many studies have shown that matching instruction to learners' stated preferences does not necessarily lead to better learning outcomes and can even hurt them (Kirschner, 2017; Pashler et al., 2008). However, preferences still shape the decisions learners make about their learning, including the widely effective learning supports, such as providing feedback, including feedback from AI systems. Given their growing importance, the need to further understand the role of learning preferences towards sources of learning support (human vs. AI) has recently become significantly more important (Peng & Wan, 2024).

Unlike traditional learning contexts where preference studies focused on instructional methods, introducing AI tutors creates a fundamentally new dynamic where learners must navigate between human and AI feedback sources. Understanding how learners perceive and interact with different feedback sources becomes crucial for designing effective learning environments that can successfully combine human and AI instruction. Such understanding may contribute to the development of more effective AI-assisted learning systems, promoting their widespread adoption and deeper integration into education. Additionally, exploring this topic is crucial for addressing the unique value of human teachers, as the question of whether AI can complement or replace certain aspects of human support has been widely debated. This investigation helps educators

identify their distinctive strengths and focus on cultivating the qualities that are irreplaceable by AI (Herawati et al., 2024; Peng & Wan, 2024).

The formation of learning preference involves a complex interaction of learners' cognition, prior experience and the learning environments (Alger & Weibull, 2019). In many situations, individuals do not really know what they prefer and they may need to construct their preference as different situations arise (Lichtenstein & Slovic, 2006). Therefore, learners' preferences would not emerge, or at least not be sufficiently constructed when choices were not provided within the real-world contexts. This phenomenon aligns with findings from the literature on self-regulated learning, which demonstrates that learners self-regulate by setting their learning goals and select and apply appropriate strategies (Pintrich, 2000; Zimmerman, 2000). To an observer, the outcomes of such self-regulation may appear to reflect their learning preferences. Consequently, when learners are presented with decision elements that are entirely unfamiliar (e.g., learning supported by technology they have never encountered before), measuring their preferences may be premature as these preferences are not yet fully formed. In some cases, individuals are able to report their preferences even without being fully informed about the alternatives beforehand, and such decisions are often based on limited information and past experiences (Johansson et al., 2012; Seidl, 2002). These theoretical insights about the constructive nature of learning preferences inform us that when measuring learners' preferences both before and after their exposure to different feedback sources could be helpful.

Algorithm aversion: Human's preference for human

Existing theories and evidence offer potential insights into understanding learners' preferences between human and AI tutors. Studies on algorithm aversion are particularly relevant to this question. The phrase 'algorithm aversion' first appeared in the work by Dietvorst et al. (2015), in which they conducted a series of experiments to show that individuals tended to prefer decisions made by humans over algorithms, even though they were aware that algorithms outperformed human decision makers in the experiment tasks.

Algorithm aversion is considered a phenomenon that exists among different contexts (Burton et al., 2020; Jussupow et al., 2020; Mahmud et al., 2022). Also, research identified some of the moderators. Those moderators include algorithms' characteristics like algorithm performance (Bogert et al., 2021; Dietvorst et al., 2015; Kim et al., 2023), algorithm transparency (Jauernig et al., 2022) and human control (Chatila et al., 2019; Dietvorst et al., 2018). Users' characteristics were also identified as moderators, including expertise in the decision task (Berger et al., 2021), familiarity with algorithms (Mahmud et al., 2022) and prior expectations of AI decisions (Turel & Kalhan, 2023). Furthermore, algorithm aversion is task-dependent, it is moderated by the characteristics of the task, including its objectivity (to what extent the task could be done with logic and computation) (Castelo et al., 2019) and complexity (Wahn et al., 2023).

Although current literature has identified the main effects and some of the boundary conditions, the reasons why people are 'inherently' averse to algorithms across various contexts remain unclear. Existing studies provide some explanations, including users' false expectations, lack of decision control, combating intuitions and conflicting concepts of rationality (Jussupow et al., 2020), but they still lack deeper insights and robust theoretical frameworks. As such, it remains unclear whether learners prefer human tutors or AI, given that the fundamental mechanism of algorithm aversion is still under-explored.

While studies on algorithm aversion are limited, we posit that status quo bias (SQB) theory could offer complementary insights to the concept of algorithm aversion with connection to AI systems in education. SQB describes an individual's preference to avoid changes

and maintain current situations (Godefroid et al., 2023; Samuelson & Zeckhauser, 1988), which has been widely observed in educational technology adoption (Anton et al., 2022; Debarliev et al., 2020; Niranga et al., 2022). While algorithm aversion describes the observed preference for human over algorithmic decisions, SQB explains the fundamental psychological mechanisms that may drive this aversion. This is especially pertinent in education where learners have developed long-term familiarity with human teachers as their primary source of feedback and guidance. This bias operates through two primary mechanisms. First, through rational decision making (Niranga & Sedera, 2023), where individuals consciously evaluate the perceived usefulness of new technologies against familiar approaches (Davis, 1989; Hart & Laher, 2015). Second, through loss aversion (Chi et al., 2020; Samuelson & Zeckhauser, 1988), where potential losses from changing the status quo are weighted more heavily than potential gains (Huang et al., 2022; Levy, 1992). In educational settings, where learning involves complex social and emotional dimensions, these mechanisms may be particularly relevant in explaining learners' preferences between human and AI tutoring systems.

Learner's preference for feedback from AI and human tutors

In educational settings, whether learners prefer feedback from an AI tutor over that from a human tutor is not entirely clear from the existing literature. As has been discussed in 2.1 section, the answer to this question, along with its underlying explanation, is of great significance. While individuals' aversion towards algorithms has been systematically investigated across contexts, evidence within the realm of educational technology is still limited. In examinations, students have been found to prefer suggestions from human experts over those from statistical models (Önköl et al., 2009). This finding from the Önköl et al. (2009) study indicates that aversion to algorithmic learning feedback might exist in human–AI interaction in learning contexts. Shao et al. (2024) explored individuals' preferences for choosing AI-based educational tools versus human educators. The result of their study revealed that users were more likely to choose human educators for their children and were more willing to share this preference with others. Peng and Wan (2024) investigated students' preferences between human and AI teaching assistants (TAs). Their findings revealed that these preferences were shaped by trust disparities (competence-based, goodwill-based and integrity-based) and moderated by factors, such as social anxiety and problem complexity. While students with higher social anxiety tended to prefer AI TAs, they were more likely to prefer human TAs when dealing with complex problems. Given the limited empirical studies and the unclear conclusions regarding whether learners prefer AI to human teachers, coupled with the complex moderating factors that influence learners' preferences, further research is needed to provide additional empirical evidence.

Studies on algorithm aversion suggest that humans may inherently prefer feedback from other humans. However, generative AI has the potential to be a game changer. Generative AI, represented by ChatGPT, as a cutting-edge technology, with its powerful text comprehension and natural language processing capabilities (Koraishi, 2023; Nguyen, 2023; Qureshi, 2023), seems to be challenging our understanding of algorithm aversion. For example, Kabir et al. (2023) found that although more than half of ChatGPT's answers to programming questions contain incorrect information and over 70% are verbose, the participants in the experiment still preferred ChatGPT's answers due to their comprehensiveness and clarity.

Nevertheless, there is also some contradictory evidence: Nazaretsky et al. (2024) examined the impact of the feedback provider's identity (AI vs. human) on students' perceptions

and found that revealing the provider's identity influenced students' preferences, leading them to prefer human-generated feedback and rate AI-generated feedback lower. Given the limited and contradictory evidence, it is still not clear if learners would prefer generative AI over human tutors. As such, we propose our first research question:

RQ1. Do learners prefer feedback given by generative AI over human tutors in educational settings?

Interaction design of generative AI-powered systems

Though we could take generative AI as potentially unique in gaining preference compared to traditional AI applications, the interaction design may also play a crucial role in shaping users' preferences (Payne et al., 2000). Previous research has demonstrated that the design of user interaction interfaces can significantly affect how users perceive and engage with AI tools, particularly in addressing algorithm aversion (Bodily et al., 2018).

Interaction modes are an essential part of interaction design. Two primary interaction modes emerge in the current application of generative AI: free-form dialogue interfaces and structured interface-guided interactions. Studies have suggested that users' reluctance to rely on AI applications could stem from a lack of trust and the perception that AI cannot match human intuition and understanding (Dietvorst et al., 2015; Prah & Van Swol, 2017). The design of interaction interfaces might either mitigate or exacerbate these concerns.

Free-form dialogue interfaces, as exemplified by ChatGPT, enable continuous, multi-turn interactions that mirror human conversation patterns. This design allows learners to engage in interactive exchanges similar to human tutoring, making the AI system more approachable and relatable. More importantly, such an interaction mode can help overcome the 'perfection schema' effect—where users expect algorithmic advice to be perfect (Madhavan & Wiegmann, 2007) and lose confidence rapidly when errors occur (Dietvorst et al., 2015). Through back-and-forth conversations, users can seek clarification, request modifications and validate the AI's responses, which has been shown to reduce algorithm aversion (Berger et al., 2021; Dietvorst et al., 2018). In educational context, this can lead to learners increasing their preferences towards AI systems.

Another commonly used interaction design—the structured interface interaction—typically provides one-way feedback triggered by users' request (e.g., clicking a button and showing results in a structured interface). Logg et al. (2019) found that users expect feedback sources to demonstrate the ability to identify and correct errors. Nevertheless, when faced with static and non-interactive feedback, users may perceive the system as fundamentally flawed and likely to repeat errors. Even when the feedback is accurate, the inability to validate through further interaction may prevent users from recognizing its value. Unlike human advisors, where imperfection is expected and users are willing to engage in multiple rounds of inquiry (Logg et al., 2019), static AI interfaces may reinforce algorithm aversion and lose learners' preferences.

Although existing research suggests that free-form dialogue could enhance learners' attitudes towards learning tools (Yan & Zhang, 2024) and generate more positive emotional responses (AlGhamdi, 2024), robust evidence is still lacking on what factors are influencing learners' attitudes and preferences. While our focus is on comparing different AI interaction modes, including human tutors as a reference point provides valuable context for understanding how these design choices affect learner preferences. Human tutoring represents the traditional benchmark for educational feedback and the standard against which new technologies are often measured. By comparing both AI interaction modes against this established reference,

we can better assess how different interface designs might help overcome or potentially reinforce algorithm aversion. As such, we proposed our second research question:

RQ2. How does the preference differ for feedback given by generative AI (free-form dialogue) compared to AI-powered writing analytic tools (structured interface) and human tutors?

METHODS

Participants

Initially, 135 participants were recruited through campus-wide advertisements at a large public university in China. The study was advertised as a reading and writing task with a base compensation of 400 RMB and a potential performance-based bonus of 50 RMB. We employed a rolling recruitment process where participants were randomly assigned to one of the four conditions upon enrolment. From an initial sample of 135 participants, our final analysis included 114 participants after removing cases with incomplete post-task data and those showing non-engagement patterns in our measurement instruments. This recruitment and data quality screening process, while resulting in unequal group sizes, does not compromise internal validity as the random assignment was maintained throughout.

The final sample consisted of 114 undergraduate students from various academic disciplines. All participants were non-native English speakers with demonstrated English proficiency through China's College English Test Band 4 (CET-4). We collected comprehensive data on participants' demographics, English proficiency and prior experience with AI tools. Detailed participant characteristics are provided in [Appendix D](#). This sample could represent high-achieving university students from non-English speaking countries, particularly those pursuing higher education in contexts where English is used as an academic language but not the primary medium of instruction.

Experiment design

[Figure 1](#) shows the overall design and process of the experiment. The experiment was conducted in a laboratory. The participants were asked to complete the task with four steps: the pre-task, reading and writing, essay revision and post-task. The writing task

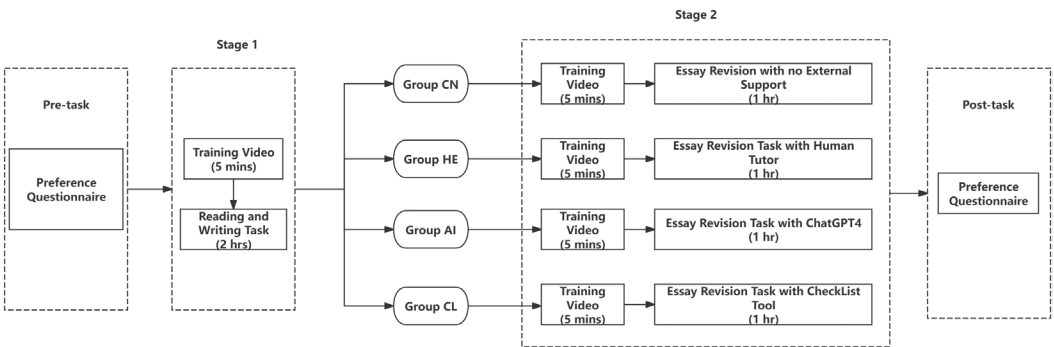


FIGURE 1 Experiment process. AI, artificial intelligence group with ChatGPT; CL, checklist group with AI-powered writing analytics tools; CN, control group; HE, human expert group.

was designed to simulate authentic academic requirements by having participants write an essay integrating multiple sources, which mirrors a common requirement in their university coursework, requiring them to write an academic essay integrating multiple sources, which is a common requirement in their university coursework.

1. *Pre-task.* During the pre-task, participants were asked to complete a questionnaire in which their preferences for human tutor and AI were measured before they went to the laboratory to complete the writing and revision tasks.
2. *Stage 1: Reading and writing.* Participants first watched training videos on using the learning tools embedded in the learning platform. These videos explained large language models (LLMs) as 'advanced AI systems trained on vast amounts of text data that can understand and generate human-like language responses', with ChatGPT introduced as an example. AI-powered writing tools were defined as 'software applications that leverage artificial intelligence to provide automated feedback on various aspects of writing, including grammar, style and content organization'. They then completed a 2-hour task involving both reading and writing. The reading materials covered *artificial intelligence, differentiation in the classroom* and *learning scaffolding*. Participants were asked to integrate these topics into a 200–400 word essay in English, envisioning education in 2035, with a provided rubric to guide their writing.
3. *Stage 2: Essay revision.* After completing the task, participants took a rest for 15 minutes before revising their essays. They were randomly divided into four groups with different types of support: (1) The Control (CN) group (27 participants) revised independently without external help. The CN group served as a baseline control, which was essential for two reasons. First, as discussed in 2.1 section, preferences are not fixed traits but could be constructed through experience in specific contexts. The writing task itself may prime learners to reflect on what kind of support they need, potentially influencing their preferences. Second, methodologically, a control group allows us to distinguish between preference changes caused by the task itself versus those resulting from different types of feedback support. (2) The Human Expert (HE) group (24 participants) received support from a human expert, who is a senior English academic editor and university instructor of academic writing. The experiment was conducted in individual sessions, with only one participant working with the human expert at a time, ensuring dedicated one-on-one support. (3) The Artificial Intelligence (AI) group (35 participants) used ChatGPT 4.0 through a free-form dialogue interface. Participants in this group were allowed to freely interact with ChatGPT in a conversational manner, asking questions, seeking clarifications and engaging in multi-turn discussions about their writing. This interaction mode mimics natural human tutor conversations, enabling dynamic and contextual feedback. (4) The Checklist (CL) group (28 participants) used a AI-powered writing analytics tool (presented as the Checklist tool in the webpage) that represents a structured interface-guided interaction mode. While this tool also utilizes ChatGPT 4.0's capabilities through API calls and prompt engineering, it presents feedback in a predetermined, one-way format. The interface provides specific categories of feedback (e.g., typos, grammar, cohesion and clarity) through a checklist-style structure. Unlike the AI group's free-form dialogue, users in this group received standardized feedback without the ability to engage in back-and-forth conversation or request immediate clarifications. Both tools in the AI group and the CL group leveraged the same underlying ChatGPT 4.0 technology, but differed fundamentally in their interaction design: one offering free-form dialogue, and the other providing structured, interface-guided feedback.
4. *Post-task.* After participants had finished the revision task, they were asked to complete the post-task. In the post-task, participants in all four conditions filled in the questionnaire again in which their preferences for AI and human tutor were rated.

Experiment environment

Our experiment utilized a Moodle-based learning platform to support the activities. In Stage 1, participants could read the materials and write the essays on the platform. In Stage 2, participants in different groups received different treatment, the treatment was based on these tools:

Generative AI tool

The AI group accessed a chat interface (Figure 2) connected to ChatGPT 4.0 through API calls. The system was configured with specific educational constraints through carefully designed prompts that: (1) prevented direct solution provision, (2) encouraged critical thinking through guided feedback, (3) maintained focus on learning objectives and (4) ensured consistent evaluation standards. The system was prompted to act as an academic writing tutor with knowledge of the task context and materials. It engaged in constructive dialogue while offering suggestions and ratings based on participants' queries.

Ask teacher tool

The HE group interacted with a human expert through a dedicated chat box named Ask Teacher Tool (Figure 3). This interface enabled real-time communication with a senior English academic writing teacher who could monitor essay revisions through the platform. The human expert followed similar guidelines to the demands embedded in the prompts given to the AI group, to provide constructive feedback rather than direct corrections.

AI-powered writing analytics tool

The CL group used AI-powered writing analytics tools (presented as the Checklist tool in the webpage) that provided structured feedback through four integrated functions (Figure 4):

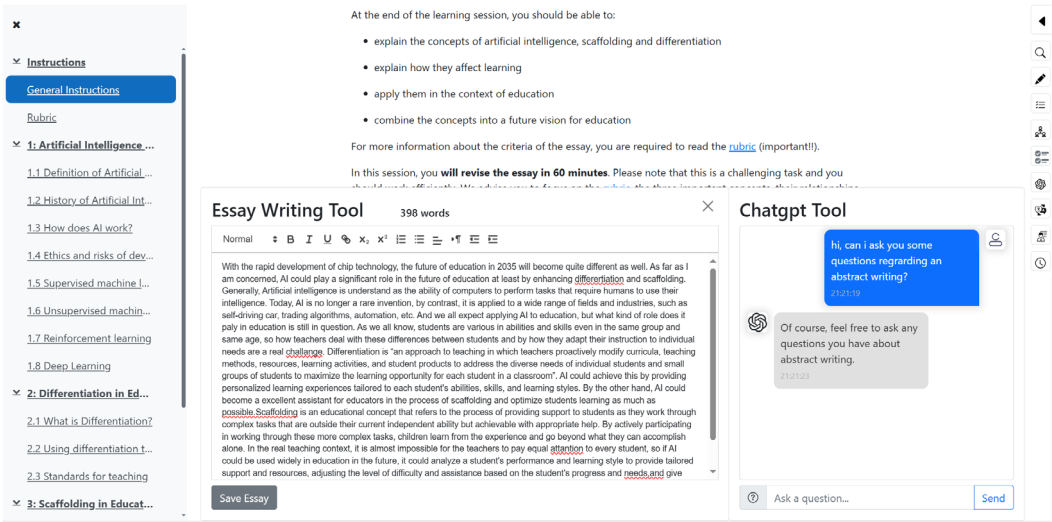


FIGURE 2 Generative artificial intelligence (AI) tool.

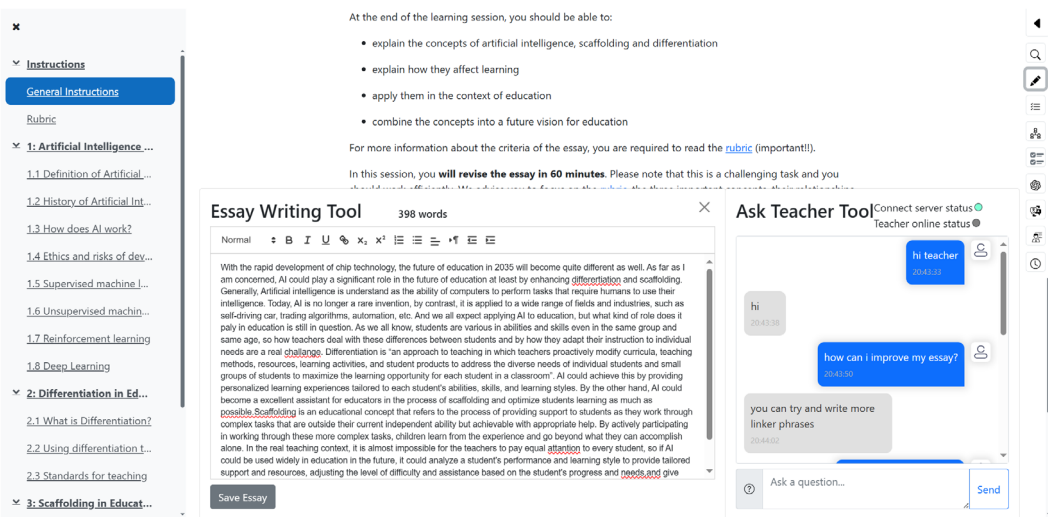


FIGURE 3 Ask teacher tool.

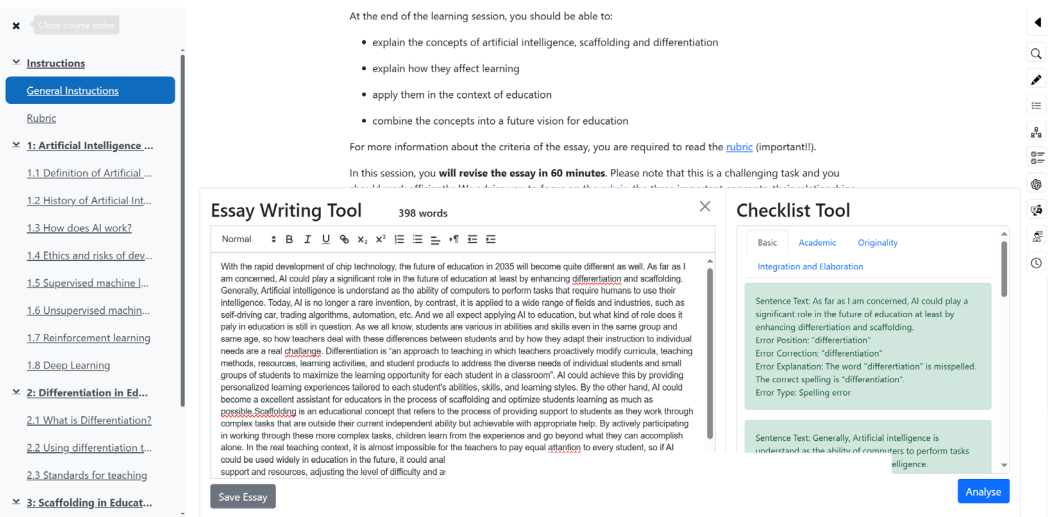


FIGURE 4 Artificial intelligence (AI)-powered writing analytic tools.

(1) basic writing analysis: identified and explained grammatical and mechanical errors through systematic text processing; (2) academic writing enhancement: analysed and suggested improvements for formal tone, structure and vocabulary based on academic writing conventions; (3) originality assessment: detected sequences of seven or more consecutive words matching source materials to ensure content authenticity; (4) integration-elaboration analysis: automatically evaluated sentences according to Bloom's taxonomy levels (Remember/Understand, Apply/Analyse, Evaluate/Create), following the approach proposed by Iqbal et al. (2024). Each function presented feedback through a colour-coded highlighting system directly within the essay text, allowing users to systematically address different aspects of their writing. Detailed technical specifications for both AI tools (generative AI Tool and AI-powered writing analytics tool), are provided in Appendix C.

Measurement of preference

As noted in 2.2 section, there are generally two approaches to operationalizing preference, either as traits or ranked choices. To ensure a robust measurement of preference, the measurement is composed of three parts: the human–AI preference scale, the human–AI preference choice and preference alteration.

Human–AI preference scale

To measure participants' preferences between AI and human expert feedback in academic writing, we developed a 5-point Likert scale grounded in established writing assessment frameworks. Our scale development was primarily based on Rezaei and Lovorn (2010)'s comprehensive writing assessment rubric, which originally evaluated five dimensions: structural organization and clarity, understanding and synthesis of argument, understanding goals and implications, support and citation and writing mechanics. From this framework, we extracted and adapted three core dimensions that represent fundamental writing abilities.

These dimensions were translated into our first three items: (1) 'Compared to AI, human tutors can better understand the main idea of my article and provide more helpful suggestions'. (2) 'Compared to AI, human tutors are more effective at identifying grammatical and spelling errors in my article'. (3) 'Compared to AI, human tutors offer superior advice for optimising arguments, structuring the text, and crafting sentences'.

Additionally, informed by recent research highlighting the crucial role of interactive feedback in writing development Imran and Almusharraf (2023), we included two items addressing communicative aspects: (4) 'Compared to AI, human tutors communicate more smoothly with me'. (5) 'Compared to AI, human tutors have a better understanding of the problems I am currently facing'. These items reflect the importance of modification-oriented dialogue in helping writers better identify and address both global and local problems in their writing (Check Appendix A for more details). The scale demonstrated acceptable internal consistency with Cronbach's alpha of 0.72 in the pre-task and 0.80 in the post-task. Notably, we did not use the term 'Generative AI' in these questions to measure learners' preferences for AI in a broader sense.

Human–AI choice

The second part of our measurement involved a dichotomous item that allowed the participants to express their general preference through a direct choice: 'Overall, if I had to choose one, I would prefer to receive help from an AI/human tutor' (Check Appendix A for more details). This method is in line with studies on algorithm aversion (Dietvorst et al., 2015, 2018), which employ trade-offs between two alternatives to gauge preference. Such a dichotomous item involved two responses. Response 0 means the participant preferred AI to a human tutor, and 1 means the participant preferred a human tutor to AI.

Preference alteration

As mentioned in 2.1 section, learners' preferences are not fixed but rather constructed through experience. Given this dynamic nature of preferences, we anticipated that participants' preferences could change after experiencing different writing support tools during the experiment. Therefore, we incorporated preference alteration as an additional measure. We assessed the

Human–AI preference choice both before (pre-task) and after (post-task) the experiment. By comparing the results of their preference choices across these two points, we could determine participants' preference alterations. Preference alteration was operationalized as participants' choice for human (1) or AI (0) in the post-task minus their choice in the pre-task. A value of -1 indicated that the participant altered their preference from human tutor to AI; a value of 1 indicated that the participant altered their preference from AI to human tutor; a value of 0 indicated that the participant's preference remained unchanged.

Data analysis

To address RQ1, we first examined the overall preference for feedback from AI versus human tutors as reported by learners both before and after the task. This analysis primarily utilized descriptive statistics. We assessed learners' preferences through two measures: the human–AI preference scale and the human–AI preference choice. For the human–AI preference scale, we calculated the mean score along with the standard deviation for the items. Given that this scale was based on a 5-point Likert format, a mean score exceeding 3 would suggest a preference for human tutors. Additionally, we compared the proportion of learners who selected AI with those who chose human tutors to identify the overall preference in terms of the human–AI preference choice.

To address RQ2, a series of ANOVAs were conducted to compare learners' preferences in terms of the human–AI preference scale, human–AI preference choice and preference alteration between the four groups. A p -value < 0.05 would indicate a significant difference among the four groups, and in such a case Tukey's HSD would be applied to adjust for pairwise comparisons and control for Type I error; an adjusted p -value smaller than 0.05 would suggest significant difference between the two conditions in pairwise comparisons'. Also, we tested learners' preference choice alteration in each of the groups by means of a paired t -test, a p -value smaller than 0.05 would suggest a significant difference in learners preference choice before and after the task.

RESULTS

In this section, we present the analysis results according to the research questions. The descriptive statistic results are shown in [Appendix B](#), including descriptive statistics results of human–AI preference score, human–AI choice score, preference alteration before and after task, as well as descriptive statistics of ChatGPT Dialogue Interactions of the AI group.

RQ1: Do learners prefer feedback given by generative AI over human tutors in educational settings?

We first analysed the data from the human–AI preference scale in the pre-test. The results showed that the participants' preference was geared towards a human tutor (mean = 3.921 , standard deviation = 0.517). Moreover, human–AI choice data also confirmed that human tutors would be preferred if participants had to trade between human tutors and AI. A total of 87.2% of participants chose the human tutor in the pre-task. Meanwhile, we measured learners' human–AI preferences and human–AI choices again after they completed the experimental task. Our results showed that participants still showed their preference for the human tutor (mean = 3.811 , standard deviation = 0.567).

The human–AI choice score showed that 86.00% of the participants chose the human tutor in the post-task as their preferred learning facilitator against an AI tutor. These results suggest the presence of algorithm aversion in this educational context. The initial preference for human tutors remained relatively stable even after participants gained experience with AI tools.

RQ2: How does the preference differ for feedback given by generative AI compared to AI-powered writing analytic tools and human tutors?

Initial group comparisons

For the pre-task, ANOVA results indicated that there was no significant difference in terms of human–AI preference score between the four groups ($F=1.188, p=0.318, \eta^2=0.030$). Also, ANOVA results showed there was no significant difference in terms of human–AI choice score between the four groups ($F=0.967, p=0.411, \eta^2=0.030$). Figure 5 shows the mean human–AI preference scores and human–AI choice scores of the four groups. The error bars represent the standard error of the mean. As the analysis of human–AI preference score and human–AI choice revealed no significant differences across the groups ($p>0.05$), the results confirmed the successful randomization and established baseline equivalence across the groups.

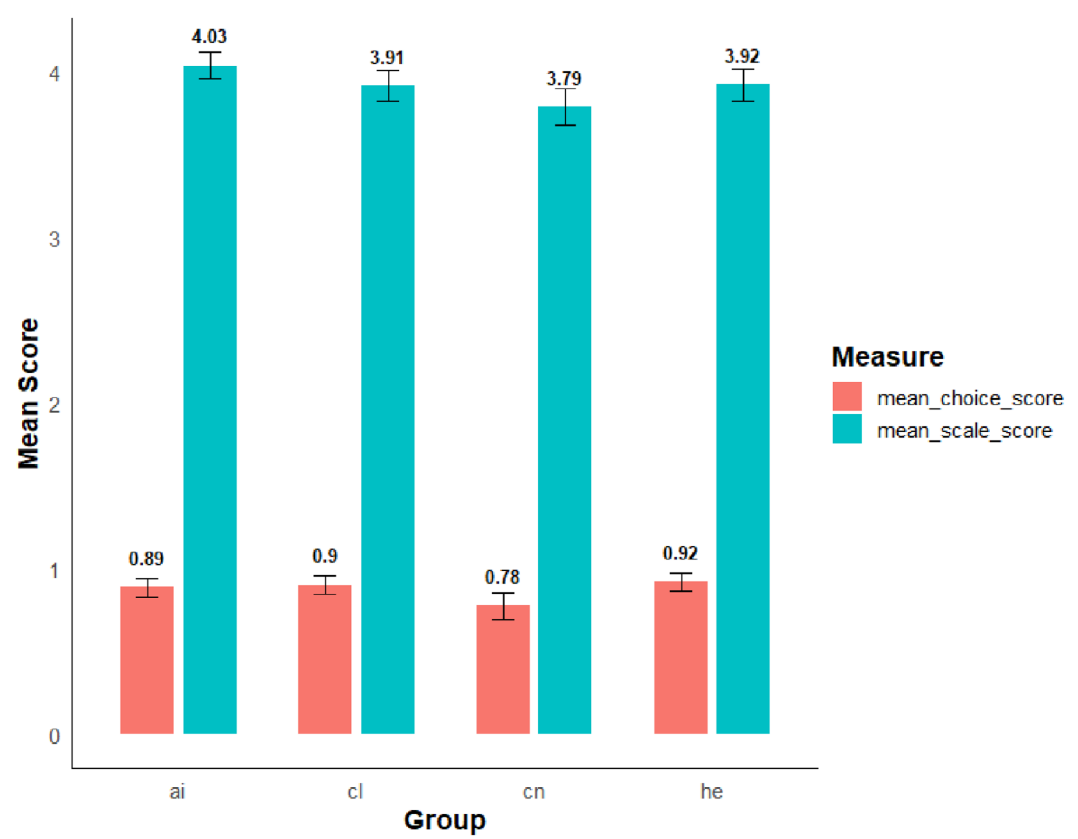


FIGURE 5 Initial preference by group.

Post-task preference patterns

The results of the ANOVA for the human–AI preference scores on post-task revealed a significant difference between the four groups ($F=7.958$, $p=0.000$, $\eta^2=0.18$). Consequently, post hoc analysis was performed using Tukey's HSD to control for Type I error (Table 1). Pairwise comparisons results indicated two significant effects (Figure 6): (1) The HE group's human–AI preference scores were significantly higher than those of the AI group (mean difference = 0.585, p -adjusted = 0.000, Cohen's $d=1.010$). (2) The HE group's human–AI preference scores were significantly higher than those of the CN group (mean difference = 0.608, p -adjusted = 0.000, Cohen's $d=1.204$).

TABLE 1 Pairwise comparisons of human–AI preference scores among the groups.

Group	Mean difference	0.95 LCI	0.95 RCI	p adj.
CL-AI	0.284	−0.061	0.629	0.144
CN-AI	−0.023	−0.371	0.326	0.998
HE-AI	0.585	0.255	0.946	0.000***
CN-CL	−0.307	−0.674	0.060	0.134
HE-CL	0.301	−0.077	0.680	0.167
HE-CN	0.608	0.227	0.990	0.000***

*** $p < 0.001$.

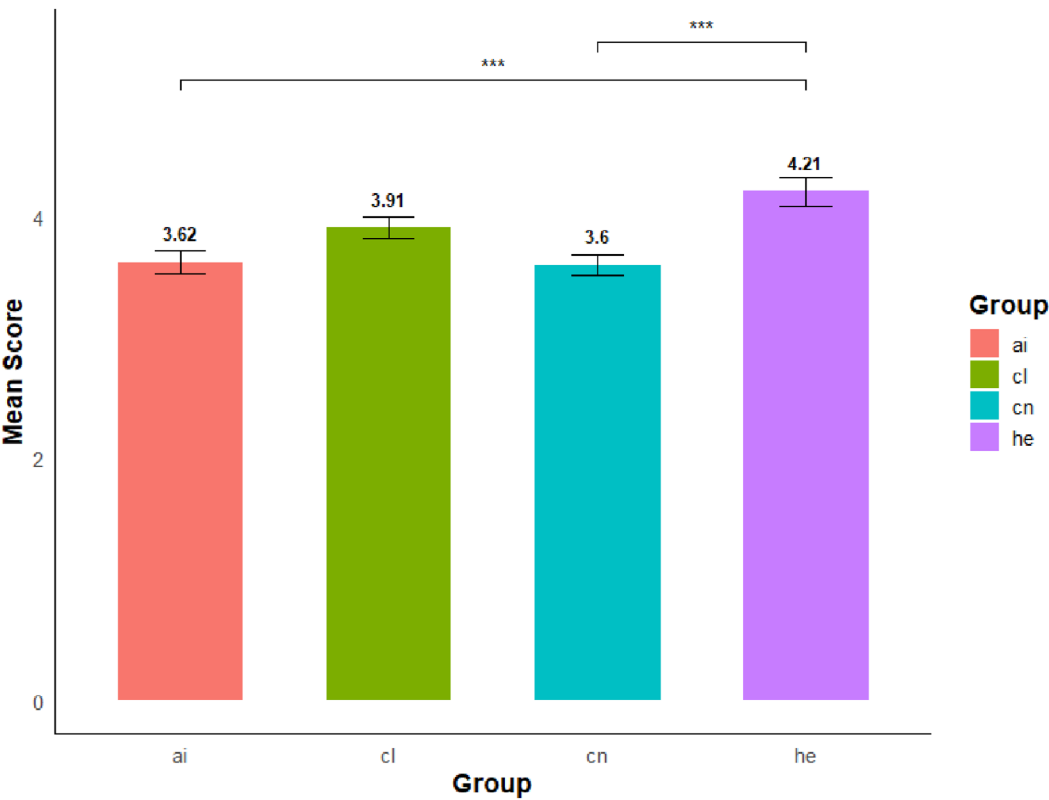


FIGURE 6 Mean human–AI preference scale scores by group after the task.

The results of the ANOVA for the choice of human–AI revealed a significant difference between the four groups ($F=4.735$, $p=0.004$, $\eta^2=0.11$) on the post-task. Post hoc analysis was performed using Tukey's HSD (Table 2), and pairwise comparison results indicated two significant effects which were different from the results of human–AI preference scores: (1) The HE group's human–AI choice scores were significantly higher than those of the AI group (mean difference = 0.286, p -adjusted = 0.008, Cohen's $d=0.807$). (2) The CL group's human–AI choice scores were significantly higher than those of the AI group (mean difference = 0.2500, p -adjusted = 0.019, Cohen's $d=0.686$).

Analysis of preference alteration

To understand how different feedback sources influenced changes in preference, we examined preference alteration scores (where positive values indicated an increase in preferences for human tutors, negative values indicated an increase in preferences for AI). The mean alteration score across all participants was -0.009 , indicating a slight overall shift towards AI preferences. However, ANOVA results revealed significant group differences ($F=3.677$, $p=0.014$, $\eta^2=0.09$). Post hoc analysis was performed using Tukey's HSD (Table 3), and pairwise comparison results indicated two significant effects (Figure 7): (1) The HE group's preference alteration scores were significantly higher than those of the AI group (mean difference = 0.255, p -adjusted = 0.036, Cohen's $d=0.576$). (2) The CL group's preference alteration scores were significantly higher than those of the AI group (mean difference = 0.243, p -adjusted = 0.037, Cohen's $d=0.585$).

Table 4 reports on the results of the analysis of the difference between participants' choice for human or AI before and after the task using paired t -test. The AI group showed a marginally significant difference ($p=0.057$) between the choice in post-task and pre-task, which indicates the participants showed a greater preference for AI after the task.

TABLE 2 Pairwise comparison of human–AI choice scores among the groups.

Group	Mean difference	0.95 LCI	0.95 RCI	p adj.
CL-AI	0.250	0.030	0.470	0.019*
CN-AI	0.101	−0.122	0.323	0.641
HE-AI	0.286	0.056	0.516	0.008**
CN-CL	−0.149	−0.384	0.085	0.347
HE-CL	0.036	−0.206	0.277	0.980
HE-CN	0.185	−0.058	0.429	0.200

** $p<0.01$; * $p<0.05$.

TABLE 3 Pairwise comparison of preference alteration scores among the groups.

Group	Mean difference	0.95 LCI	0.95 RCI	p adj.
CL-AI	0.243	0.010	0.476	0.037*
CN-AI	0.208	−0.027	0.444	0.101
HE-AI	0.255	0.011	0.498	0.036*
CN-CL	−0.034	−0.282	0.213	0.984
HE-CL	0.012	−0.244	0.267	0.999
HE-CN	0.046	−0.211	0.304	0.966

* $p<0.05$.

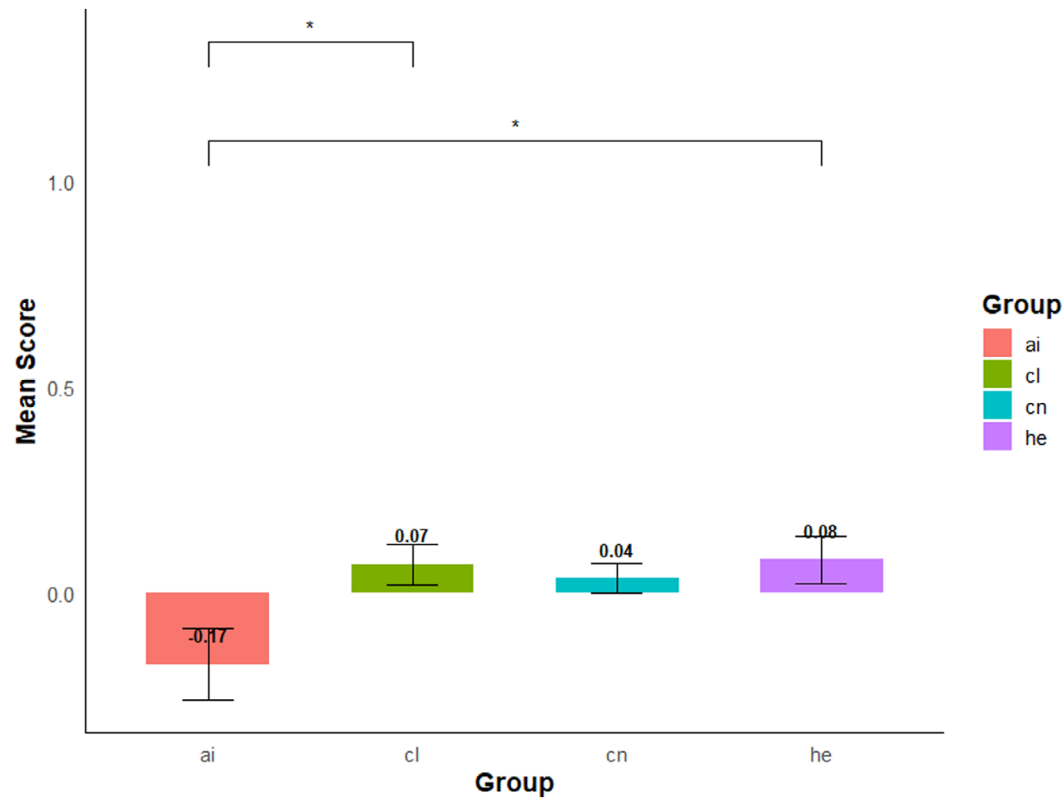


FIGURE 7 Mean preference alteration scores by group.

TABLE 4 Comparison of pre-task and post-task preference choice scores within groups.

Group	Mean difference	<i>t</i>	<i>df</i>	<i>p</i>	Cohen's <i>d</i>
AI	−0.171	1.974	34	0.057*	0.432
CL	0.071	1.441	27	0.161	0.275
CN	0.037	1.000	26	0.327	0.090
HE	0.083	1.446	23	0.162	0.417

Note: The direction of the mean difference is post-task choice score minus pre-task choice score, smaller value indicates an alteration of preference from human tutor to AI.

* $p < 0.1$.

Based on the above results, we found an interesting phenomenon from our experiment: Participants in the HE group preferred human tutors more after the task, and participants in the AI group preferred AI more after the task. But surprisingly, participants in the CL group showed the same pattern of preference alteration as those in the HE group. As shown in Figure 8 and Table 2, both the HE and CL group reported a significantly higher human–AI choice score for human tutors, while there was no significant difference between the HE and CL group in terms of human–AI choice. Results of preference alteration show the same effect: Both the HE and CL group showed significantly higher preference alterations to human tutors than the AI group did, as is shown in Figure 8 and Table 3.

For the AI group, we found that the participants' preferences for AI increased after the task ($p = 0.057$), with a moderate effect size (Cohen's $d = 0.432$). This suggests a meaningful shift

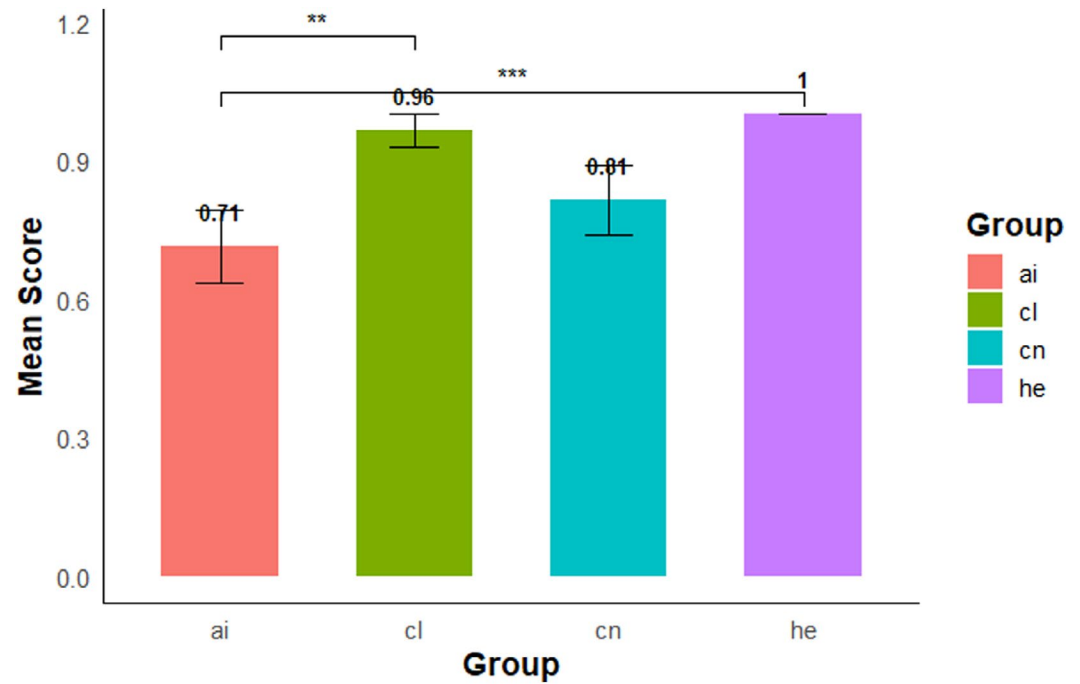


FIGURE 8 Mean human–AI preference choice scores by group after the task.

in preference towards AI following direct interaction experience. For the HE and CL groups, while the pre/post-changes did not reach statistical significance, several lines of evidence suggest meaningful preference changes. First, the effect sizes were substantial: the HE group showed a moderate effect size (Cohen's $d=0.417$) and the CL group showed a small to moderate effect size (Cohen's $d=0.275$), both indicating practically meaningful changes in preference towards human tutors. Second, the statistical insignificance likely resulted from a ceiling effect due to high baseline preferences—for instance, even a simulated value of 100% choice score in the CL group's post-task would only yield a p -value of 0.08. These effect sizes are particularly noteworthy when compared to the CN group's minimal effect size (Cohen's $d=0.090$), suggesting that both human tutor and structured AI interactions did influence participants' preferences. The stability of the CN group's preferences throughout the experiment provides a useful reference point, highlighting that the preference changes in other groups were likely due to their specific feedback experiences rather than the task itself.

Individual factors influencing preferences

While our primary analysis focused on group-level effects, regression analysis revealed that individual characteristics also influenced learners' preferences for different feedback sources. Before the experimental task, age showed a marginally significant effect, with older students demonstrating a slightly higher preference for AI tutors, and prior writing tools experience was associated with a higher preference for human tutors. After experiencing the different feedback sources, English proficiency emerged as a significant factor, with higher-proficiency students showing an increased preference for AI tutors.

Further analysis of interaction effects revealed important moderation patterns: Learners with higher English proficiency demonstrated stronger preferences for human tutors

specifically when using the structured interface (CL group); learners with more experience using large language models showed relatively lower preference for human tutors in the control group; and those with more writing tool experience exhibited relatively lower preference for human tutors when using the structured interface. These findings suggest that learners' language proficiency and technological experience are important factors influencing their feedback source preferences, highlighting the need for personalized approaches to educational technology implementation.

It is important to note that these findings represent preliminary observations from our sample. Given the relatively modest sample size and the exploratory nature of these analyses, we present these results as promising directions rather than definitive conclusions. A more comprehensive investigation of these individual factors would require larger samples and more targeted experimental designs. Complete regression tables and detailed statistical analysis can be found in [Appendix E](#), which may serve as a foundation for future research specifically focused on individual differences in feedback preferences.

DISCUSSION

The overall persistence preference for human tutors

Addressing RQ1, our results showed a strong initial preference for human tutors, with 87.2% of participants choosing human over AI support. As discussed in our theoretical framework (Section 2.1), preference is not a fixed trait but rather constructed through experience and interaction. Therefore, we examined whether initial preferences would change after participants gained actual experience with different tutoring approaches. The pre-task preference might have limitations since many participants had limited prior exposure to learning tools and generative AI, or unclear expectations about human tutor assistance. However, the post-task results reinforced the strong preference for human tutors, with a mean preference score of 3.81, and 86.0% of participants still choosing human tutors. This preference was particularly pronounced in the CL group (96.0%) and HE group (100%). Even in the AI group, which showed the lowest preference for human tutors, 71.0% still preferred human support. These consistent findings across both pre- and post-task measurements provide robust evidence that learners maintained their preference for human tutors in academic writing tasks, with this preference becoming even stronger after experiencing human tutor support.

Our findings align with previous research on algorithm aversion, where individuals consistently prefer human over algorithmic decision making across various contexts (Dietvorst et al., 2015; Jussupow et al., 2020). The persistence of this preference can be understood through both algorithm aversion (Dietvorst et al., 2015) and status quo bias frameworks (Godefroid et al., 2023; Samuelson & Zeckhauser, 1988). As college students have primarily experienced classroom-based education with human teachers, this familiarity forms their status quo, making them more inclined to maintain existing patterns of learning support. This tendency operates through the two mechanisms we discussed earlier. First, through rational decision making, learners consciously evaluate the perceived usefulness of different tutoring approaches. Human tutors are often viewed as more capable of understanding complex learning needs (Shemshack & Spector, 2020), providing nuanced feedback (Chi et al., 2001) and adapting their teaching strategies in real-time (VanLehn, 2011). These perceived advantages align with learners' prior positive experiences with human teachers and their understanding of effective learning support. Second, through loss aversion, learners may overweight potential losses in switching from human to AI tutors. These perceived losses include reduced emotional support, decreased social presence and limited ability to build rapport—elements that learners consider crucial for effective learning experiences (Huang et al., 2022). In educational

contexts, where learning often involves complex emotional and social dimensions, these concerns about potential losses become particularly salient.

Our findings on learners' preference for human tutors can be further explained. In our previous qualitative investigation using the same experimental setup (Shen et al., 2025), we identified that learners prioritize five core values: responsiveness, social comfort, autonomy, freedom from bias and privacy. Our interview data revealed that human tutors excelled in 'responsiveness', accurately interpreting learners' intentions and emotions even when poorly expressed. In contrast, ChatGPT demonstrated advantages in 'social comfort', with our participants reporting 'minimal sense of guilt or shame' and reduced social pressure, while also supporting 'autonomy' by offering learners 'greater power of selectivity and decision making'. However, our analysis of value trade-offs showed that despite AI providing a judgement-free environment and decision freedom, learners still prioritized the experience of being deeply understood.

The role of interaction design: Mitigating algorithm aversion through free dialogue

Both the AI and CL groups were supported by ChatGPT, yet the AI group gained more preference for AI after the task, while the CL group not only failed to do so but even showed a shift in participants' preference towards human tutors. This raises the question: Why did the AI group gain more preference for AI, while the CL group saw a shift towards a preference for human tutors just like the HE group did? Our findings indicate that learners' preferences were not solely fixed on human tutors but could be influenced by the characteristics of the AI applications in use (i.e., their interaction modality). The preference alteration observed in the AI group suggests that free-dialogue interfaces could effectively mitigate algorithm aversion through multiple potential mechanisms.

First, free-dialogue creates a more natural and human-like interaction environment that can help overcome algorithm aversion by offering room for feedback correction. When learners can freely express their questions and concerns in natural language, the AI application is more approachable and less 'algorithmic' in learners' perception. More importantly, through continuous dialogue, learners can question, challenge and seek clarification, which helps break the 'perfection schema' effect—where users expect algorithmic advice to be perfect and lose confidence rapidly when errors occur (Dietvorst et al., 2015; Madhavan & Wiegmann, 2007). The ability to engage in natural conversation, respond contextually and maintain dialogue coherence also enables AI to exhibit qualities that learners associate with human intuition and understanding, addressing a key source of algorithm aversion (Dietvorst et al., 2015; Prah & Van Swol, 2017).

Second, the utility alignment with diverse learning needs likely plays a crucial role. While the structured interface was limited to specific functions, free-dialogue allowed learners to seek feedback for any aspect of their writing (Kim et al., 2024). When learners' needs extended beyond the predetermined categories offered by the AI-powered writing analytics tools, the system's utility became limited. The free-dialogue interface's ability to address unexpected or complex learning needs through flexible interaction might explain why learners found it more valuable, despite using less sophisticated prompts.

Interestingly, these advantages of free-dialogue persisted even though the structured interface used carefully designed and tested prompts developed by researchers (see Appendix C for the complete prompt engineering details). In contrast, students in the free-dialogue condition had to create their own prompts, which were typically more casual and less optimized (e.g., 'Can you check my grammar?' or 'Is this paragraph clear?'). This suggests that the benefits of flexible interaction and comprehensive utility might outweigh the advantages of technically refined prompts.

These findings suggest that the path to overcoming algorithm aversion in AI systems in education may lie not just in improving algorithmic performance, but in designing interaction modes that both emulate human-like communication patterns and provide flexible support for diverse learning needs. This has important implications for the design of AI systems in education, suggesting that developers should focus not only on technical optimization but also on creating natural, flexible interaction experiences that can comprehensively support learning.

Reflecting on our findings through the lens of our companion studies (Chen et al., 2025; Fan et al., 2025), process mining analysis revealed that AI group learners engaged in significantly fewer metacognitive processes, essentially 'outsourcing' their self-regulation to ChatGPT through frequent interaction loops. This cognitive offloading phenomenon presents a compelling paradox: While it may mitigate algorithm aversion by making learning feel effortless, it potentially undermines the very metacognitive skills essential for deep learning. Interestingly, although the AI group outperformed the CL group in immediate task performance, there were no significant differences in knowledge transfer outcomes between groups—suggesting that the preference for free-dialogue interfaces might reflect convenience rather than deeper learning benefits. This raises a concerning possibility that invites us to question whether increased preference for AI tools should always be our design goal, or whether we should instead prioritize interfaces that maintain the productive cognitive struggle necessary for meaningful learning.

Implications

This study extends the theory of algorithm aversion (Dietvorst et al., 2015; Jussupow et al., 2020) to educational settings, demonstrating that learners generally prefer human over AI tutors, even after experiencing AI feedback. This preference pattern could potentially be understood through both rational decision-making processes (Davis, 1989; Hart & Laher, 2015) and cognitive biases (Godefroid et al., 2023; Samuelson & Zeckhauser, 1988). Though further research would be needed to empirically verify these theoretical explanations. Moreover, our findings reveal that this aversion can be effectively mitigated in educational contexts through appropriate interaction design.

For educational technology developers, our findings suggest that overcoming algorithm aversion requires more than just technical optimization. While improving algorithmic performance remains important, equal attention should be paid to designing natural and flexible interaction interfaces. Specifically, AI systems in education should prioritize natural dialogue capabilities that allow learners to express their needs freely and receive contextual responses. This approach appears more effective than structured, function-specific interfaces, even if the latter are more technically optimized. Meanwhile, AI systems should be designed to address diverse learning needs rather than focusing solely on specific functions. The ability to provide flexible support appears to be a key factor in gaining learner acceptance.

However, our findings come with an important caution: while generative AI may overcome algorithm aversion, increased preference could foster problematic learning behaviours. Research suggests students may develop 'metacognitive laziness', using AI as shortcuts rather than learning tools. This raises questions about whether AI preference actually improves learning outcomes.

Limitations and future research direction

Our study's limitations include sample size constraints that affected statistical power, particularly when comparing pre/post-task preferences where high baseline preference for

human tutors left minimal room for significant increases. We lack comprehensive insights into the mechanisms behind learners' persistent human tutor preference despite identifying interaction design's influence. The absence of standardized usability assessments limited our understanding of how interface design elements (colour-coding and presentation format) potentially affected cognitive load and engagement. Also, another limitation of our study is the absence of comprehensive process data that would allow for methodological triangulation between stated preferences and objective usage patterns. While we collected basic interaction logs, these data alone are insufficient for understanding how preferences form and evolve during the learning process. Additionally, while our preference measurement was adapted from an established writing assessment framework to ensure content validity, we recognize that additional validation studies would be beneficial to fully establish the construct validity of our adapted instrument in the specific context of comparing human and AI feedback preferences.

Regarding external validity, while our task simulated authentic academic writing, laboratory conditions may have influenced participants' approach compared to natural learning environments. Our experimental design featuring one-to-one interactions between participants and feedback sources, while allowing for controlled comparison, could benefit from more authentic configurations that better simulate real educational environments. Also, the uneven group sizes potentially affected statistical validity despite our robust analysis approaches.

Future research should employ balanced experimental groups, validate preference measurement instruments and utilize mixed methods (interviews, think-aloud protocols) to explore preference formation mechanisms. Studies should investigate whether these preference patterns persist across different subject areas, educational levels and cultural contexts.

Meanwhile, future research should implement multiple data collection methods (such as real-time preference indicators, think-aloud protocols) to triangulate between what learners say they prefer and how they actually engage with different feedback sources. Such triangulation would provide a more robust understanding of the complex relationship between interaction experiences and preference formation.

Building on our preliminary findings regarding individual factors, future studies should systematically examine how learner characteristics, such as language proficiency, prior AI experience and technological familiarity moderate preferences between human and AI feedback sources. This would provide a more nuanced understanding of how to personalize educational technology implementations to meet diverse learner needs.

Importantly, researchers should examine potential negative consequences of increased AI preference, particularly whether engaging interfaces might inadvertently promote over-reliance on AI support. This research direction could identify design principles that balance making AI systems appealing while fostering independent learning capabilities, recognizing that learner preferences don't necessarily align with optimal learning outcomes.

CONCLUSIONS

This study makes several contributions to our understanding of learners' preferences in AI-enhanced education. First, we confirm the existence of algorithm aversion in educational settings, demonstrating that learners generally prefer human over AI tutors. However, our findings also reveal that this preference is not immutable, and the interaction design of AI applications plays a crucial role through multiple mechanisms.

The free-dialogue interface of ChatGPT showed unique potential in gaining learner acceptance through three potential aspects. First, it creates a more natural and human-like interaction environment that helps overcome algorithm aversion by offering room for feedback correction. Through continuous dialogue, learners can question, challenge and

seek clarification, which helps break the 'perfection schema' effect (Dietvorst et al., 2015; Madhavan & Wiegmann, 2007). Second, free-dialogue better aligns with diverse learning needs, allowing learners to seek feedback for any aspect of their writing. Third, these advantages persisted even though the structured interface used more sophisticated prompts, suggesting that the benefits of flexible interaction might outweigh technical optimization. It is important to explicitly acknowledge, however, that interaction mode was not the sole differentiating factor in our study, as it encompasses multiple dimensions, including feedback presentation, perceived user agency and the cognitive processes engaged—aspects that collectively influenced learners' preference patterns and warrant further systematic investigation.

While human tutors remain preferred overall, our findings suggest that algorithm aversion in educational contexts can be mitigated through appropriate interaction design. This points towards a future where AI and human tutoring can be effectively integrated to enhance learning experiences, with interface design serving as a crucial bridge between algorithmic capabilities and learner acceptance. For practice, educational technology developers should focus on creating interfaces that enable flexible, mutual interactions rather than solely pursuing technical performance improvements.

However, we still need to take this consideration: learners' preferences may inadvertently conflict with their cognitive growth, as seductively engaging AI interfaces risk fostering a subtle dependency that diminishes the valuable struggle inherent in authentic learning—reminding us that educational technology must delicately balance immediate appeal with the enduring value of intellectual self-reliance.

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CONFLICT OF INTEREST STATEMENT

The authors report that there are no conflicts of interest related to the paper.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available upon request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

ETHICS STATEMENT

We obtained the ethics approval for this research project from the relevant institutional ethics committee. We also have taken steps to protect participants (learners), ensuring they were not disadvantaged and that the data have been anonymized.

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